Introduction to Python

Lab Activity & Google Colab Tutorial

Subject: ICT | Semester: 1st | Level: Beginner | Duration: 60 minutes

Student Name	Roll Number	Date	Section

Learning Objectives

By the end of this lab you will be able to:

- Write and run Python programs in Google Colab
- Use variables, data types, and print()
- Perform basic arithmetic and string operations
- Take user input with input() and display formatted results
- Apply if / elif / else decision structures

Time Plan

Time	Activity	What you will do
0 – 10 min	**Setup & Intro**	Open Colab, create notebook, learn the interface
10 – 25 min	**Tasks 1 & 2**	Hello World, variables, arithmetic, data types
25 – 45 min	**Tasks 3 & 4**	User input, f-strings, if/elif/else decisions
45 – 60 min	**Review & Submit**	Test programs, save, download or share notebook

Part 1 — Getting Started on Google Colab

Follow these steps before writing any Python code.

Step A — Opening Colab and creating a notebook

1. Open your browser and go to: colab.research.google.com
2. Sign in with your Google / Gmail account.
   Create a free account first if you do not have one.
3. Click "+ New notebook" in the bottom-left of the welcome dialog.
   A new tab opens with an empty notebook called Untitled0.ipynb
4. Rename it: click "Untitled0" at the top, type Python_Lab_YourName and press Enter.
5. Click "Connect" in the top-right and wait for the green tick.
   RAM and Disk bars appear — Colab is ready.

Step B — Understanding the interface

A "+ Code" button — adds a code cell where you type and run Python.

B "+ Text" button — adds a notes cell for headings and plain-English comments.

C Play button (triangle) on each cell — runs that cell. Output appears below.
Green tick + [1] means the cell ran successfully.

D Key shortcuts: Shift+Enter runs cell & moves on. Ctrl+Enter runs & stays. Ctrl+M B adds cell below.

> **Important — input() in Colab**
> When your program uses input(), the text box appears at the VERY TOP of the browser window — not below the cell. If the cell shows a spinning circle, scroll to the top to find the waiting input prompt.

Part 2 — Guided Tasks (Follow Along)

Read each starter code carefully, type it into a code cell in Colab, run it, then complete the to-do items.

Task 1 Hello, World! — Your first program 10 min

Concepts: print() | strings | comments

Starter code — type this in a code cell:

```
# Task 1 — Hello World
print("Hello, World!")
print("My name is Ahmad")
print("I am an ICT student")
```

How to run in Colab:

1. Click "+ Text" and type: ## Task 1 — Hello World as a heading.
2. Click "+ Code", type the starter code above, then press Shift+Enter.
   Output appears below the cell.
3. Change "Ahmad" to your own name and run again.

To-do items:

- Add a 4th print() line showing your class/section.
- Add a comment (starting with #) above each print() explaining what it does.

Bonus Challenge
Print your full name using three separate print() calls — one word each. Then combine all three in a single print() using commas.

Task 2 Variables & Simple Maths 15 min

*Concepts: variables | int | float | arithmetic operators (+ - * // %)*

Starter code:

```
student_name = "Sara"
age    = 16
height = 1.62      # float
print(student_name, "is", age, "years old")
print("Type of age:", type(age))

a = 20 ; b = 6
print("Sum:", a + b)
print("Quotient:", a // b)
print("Remainder:", a % b)
```

To-do items:

- Create variables for your name, age, and favourite subject. Print them in one sentence.
- Write a program that stores two numbers and prints sum, difference, product, and quotient.
- Calculate and print the area of a rectangle: width = 8, height = 5.
- Print the data type of each variable you created using type().

Bonus Challenge
Calculate the average of five test scores stored in five variables. Print the result rounded to 1 decimal place using round().

Task 3 User Input — Personal Info Calculator 15 min

Concepts: input() | int() / float() conversion | f-strings

Remember
When input() runs in Colab, scroll to the TOP of the browser to find the input box.

Starter code:

```
name   = input("Enter your name: ")
age    = int(input("Enter your age: "))
height = float(input("Enter height in cm: "))
height_m = height / 100
print(f"Hello {name}! You are {age} years old.")
print(f"Your height is {height_m:.2f} metres.")
```

To-do items:

- Ask the user for name, age, and height in cm.
- Convert height to metres and store in a new variable.
- Print a personal profile card using f-strings.
- Calculate and print the year the user will turn 18 (use current year 2026).

Bonus Challenge
Ask for two numbers and print all four operations (+, -, x, /) each on a separate line with a clear label.

Task 4 Decision Maker — if / elif / else 15 min

Concepts: if / elif / else | comparison operators (== != > < >= <=) | modulo trick

Starter code:

```
number = int(input("Enter a number: "))
if number % 2 == 0:
    print(number, "is EVEN")
else:
    print(number, "is ODD")

if number > 0:
    print("Positive")
elif number < 0:
    print("Negative")
else:
    print("Zero")
```

To-do items — each in a separate code cell:

- Marks grader: ask for marks (0–100) and print Distinction (>=80), Merit (60–79), Pass (40–59), or Fail (<40).

- Temperature checker: ask for temperature in °C and print Hot (>35), Warm (20–35), Cool (10–19), Cold (<10).
- Number comparison: ask for two numbers and say which is larger, or if they are equal.

Indentation Rule
Lines inside if / elif / else must be indented 4 spaces (one Tab). Colab adds this automatically after a colon. Never mix tabs and spaces.

Bonus Challenge
Login checker: ask for username and password. If both match "admin" and "1234" print "Access granted", otherwise print "Access denied".

Part 3 — Student Lab Work (Do It Yourself)

Write your own programs below without looking at the starter code. Use the lined boxes to write your code, then run it in Colab and record the output.

Instructions
1. Write your code in the green-lined boxes below. 2. Type it into a new code cell in Colab and run it. 3. Write what the program actually printed in the Output box. 4. Answer the reflection questions at the end of each exercise.

Exercise 1 — My Introduction Program

Write a program that prints: your full name, your age, your city, and your favourite hobby — each on a separate line. Add a comment above every print() statement.

Your code:

CODE

```
# Write your Python program here...
```

Output (what printed on screen when you ran it):

Expected / Actual Output:

Q1. What does the # symbol do in Python? Why is it important to add comments?

Q2. What happens if you forget the quotation marks around a string? Try it and write what error appeared.

Exercise 2 — Simple Calculator

Store two numbers in variables num1 = 45 and num2 = 12. Calculate and print: their sum, difference, product, integer quotient, remainder, and the result of num1 to the power of 2 (use ** operator). Label each result clearly.

Your code:

CODE

```
# num1 = ... # num2 = ...
```

Output:

Expected / Actual Output:

Q3. What is the difference between / (division) and // (integer division)? Give an example with numbers.

Q4. What does the % (modulo) operator return? What is 17 % 5? Calculate it by hand first, then check with Python.

Exercise 3 — Personal Profile with Input

Ask the user to enter: their name, their age, and their monthly pocket money in rupees. Then print a profile that shows: name, age, age in 5 years, yearly pocket money (monthly x 12), and daily pocket money (monthly / 30 rounded to 2 decimal places).

Your code:

CODE

```
# Use input() and f-strings...
```

Output (run with your own values):

Expected / Actual Output:

Q5. Why must you write int(input(...)) instead of just input(...) when asking for a number?

Q6. Write the f-string syntax that would print: Hello Ali! You are 16 years old. (fill in the blanks) print(f"____")

Exercise 4 — Electricity Bill Calculator

Ask the user to enter the number of units of electricity used this month. Calculate the bill using these rates: 0–100 units = Rs. 5 per unit | 101–300 units = Rs. 10 per unit | Above 300 units = Rs. 20 per unit. Print the total bill with a clear message. (Hint: use if / elif / else)

Plan your logic first (write in plain English):

PLAN

e.g. If units are 0-100 then multiply by 5...

Your code:

CODE

```
# Electricity bill calculator units = int(input(...))
```

Test your program — fill in this table:

Units entered	Expected bill	Your program printed
80 units	Rs. 400	
150 units	Rs. 1500	
350 units	Rs. 7000	
50 units	Rs. 250	

Q7. Did all four test cases give the correct answer? If any failed, what was the problem and how did you fix it?

Exercise 5 — Challenge: Student Report Card *(Optional — for fast finishers)*

Ask the user to enter marks for 3 subjects: Maths, English, and Science (each out of 100). Calculate the total and average. Print a full report card showing each subject mark, the total, the average, and the final grade based on the average (Distinction / Merit / Pass / Fail). Format the output neatly using f-strings.

Your code:

CODE

```
# Ask for 3 subject marks, calculate total and average, print report card...
```

Output (paste or write what your program printed):

Expected / Actual Output:

Q8. What new concepts or ideas did you use in this challenge that were not in the starter tasks?

Part 4 — Saving & Submitting Your Work

Saving your notebook

1. Colab auto-saves to Google Drive every few minutes. Press Ctrl+S to save manually.
   Your notebook is in: My Drive > Colab Notebooks
2. To download as a .py file: File > Download > Download .py
   Rename it: Python_Lab_YourName_RollNumber.py
3. To share with teacher: click "Share" > enter teacher's Gmail > set to Viewer > Send.
   Or copy the link and share via class portal or WhatsApp.

Submission Checklist

	Item	Marks	Given
☐	All 4 guided tasks completed with no errors	/ 5	
☐	Exercise 1 — Introduction program works correctly	/ 5	
☐	Exercise 2 — Calculator with all 6 operations	/ 5	
☐	Exercise 3 — Input & f-strings program	/ 5	
☐	Exercise 4 — Electricity bill with correct if/elif/else	/ 5	
☐	Reflection questions answered thoughtfully	/ 5	
☐	Code has comments and student name in every cell	/ 5	
☐	Exercise 5 (Challenge) — Report card program	/ 5	

Total: ________ / 95 + 5 bonus

Quick Error Reference
SyntaxError — missing colon (:) after if/else/for, or mismatched quotes. NameError — variable name misspelled or used before being defined. IndentationError — lines inside if/else must be indented exactly 4 spaces. TypeError — arithmetic on a string; use int() or float() to convert input(). ValueError — typed text when a number was expected; re-run and enter a number.

Good luck! Every programmer makes mistakes — just read the error and fix it one step at a time.

MARKS CALCULATION:

Formula:

$$\text{Weighted Marks} = \left(\frac{\text{Obtained Marks}}{\text{Total Marks}}\right) \times \text{Weightage}$$

Given:

- Obtained Marks = 95
- Total Marks = 100
- Weightage = 2.5

Calculation:

$$\left(\frac{95}{100}\right) \times 2.5$$

Result:

$$= 0.95 \times 2.5 = 2.375$$